Omri Ben-Dov

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Professional Summary

PhD student at the Max Planck Institute for Intelligent Systems working on algorithmic collective action in machine learning. My research studies how coordinated human behavior interacts with learning algorithms, with publications at ICML, WACV, and NeurIPS workshops. I also develop stable training frameworks for generative and adversarial models. Broad interests include ML-society interactions and learning under strategic behavior.

Research Experience

Max Planck Institute for Intelligent Systems — **Tübingen, Germany**
PhD student — *2023–Present, expected August 2026*
Human Aspects of Machine Learning Group: 2023–Present
Supervisors: *Samira Samadi* (MPI-IS) and *Amartya Sanyal* (University of Copenhagen).

- Focus on algorithmic collective action, studying how coordinated groups of individuals influence machine learning algorithms.

Perceiving Systems Department: 2021–2023
Supervisors: *Michael J. Black* (MPI-IS), *Silvia Zuffi* (IMATI-CNR) and *Sergi Pujades* (INRIA).

- Developed articulated 3D models of laboratory rats for pose estimation and motion analysis, combining optimization and learning-based methods for animal tracking and reconstruction.

Research Visits:

- Amartya Sanyal, University of Copenhagen (2024): Applying collective action to influence fairness in machine learning.
- Luiz Chamon, École Polytechnique (2025): Solving the dual problem to minimize a symmetric divergence.

Weizmann Institute of Science — **Rehovot, Israel**
Research Assistant — *2021*
Supervisor: *Barak Zackay*.

- Applied signal processing and matched filtering on gravitational waves data to detect binary black holes.

The Hebrew University of Jerusalem — **Jerusalem, Israel**
MSc in Micro-Flight Group — *2018–2021*
Supervisor: *Tsevi Beatus*.

- Developed an algorithm to estimate kinematic parameters of free-flying flies from video using a 3D model.

Publications and Preprints

- **Omri Ben-Dov**, Luiz F. O. Chamon (2025). **Adaptive Symmetrization of the KL Divergence**. arXiv: 2511.11159.
- **Omri Ben-Dov**, Samira Samadi, Amartya Sanyal, Alexandru Ţifrea (2025). **Fairness for the People, by the People: Minority Collective Action**. ***Oral Presentation in the Algorithmic Collective Action workshop, Neurips 2025***. arXiv: 2508.15374.
- **Omri Ben-Dov**, Jake Fawkes, Samira Samadi, Amartya Sanyal (2024). **The Role of Learning Algorithms in Collective Action**. In: *Proceedings of the 41st International Conference on Machine Learning*. Vol. 235. PMLR, pp. 3443–3461.
- **Omri Ben-Dov**, Pravir Singh Gupta, Victoria Abrevaya, Michael J. Black, Partha Ghosh (2024). **Adversarial Likelihood Estimation With One-Way Flows**. In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 3779–3788.
- Roni Maya, Noam Lerner, **Omri Ben-Dov**, Arion Pons, Tsevi Beatus (2023). **A Hull Reconstruction–Reprojection Method for Pose Estimation of Free-Flying Fruit Flies**. In: *Journal of Experimental Biology* 226.21.
- Arion Pons, Illy Perl, **Omri Ben-Dov**, Roni Maya, Tsevi Beatus (2023). **Solving the Thoracic Inverse Problem in the Fruit Fly**. In: *Bioinspiration & Biomimetics* 18.4, p. 046002.
- **Omri Ben-Dov**, Tsevi Beatus (2022). **Model-Based Tracking of Fruit Flies in Free Flight**. In: *Insects* 13.1018.

Education

Max Planck Institute for Intelligent Systems — **Tübingen, Germany**
PhD in Computer Science, Expected Graduation: August 2026 — *2021–Present*

The Hebrew University of Jerusalem — **Jerusalem, Israel**
MSc in Computer Science — *2018–2021*
Thesis: "Automatic Pose Estimation of Free-Flying Fruit Flies".
Magna cum laude.

The Hebrew University of Jerusalem — **Jerusalem, Israel**
BSc in Physics and Computer Science — *2015–2018*
Magna cum laude.

Work Experience

Intel — **Petah Tikva, Israel**
QA Automation Engineer — *2015*
- Developed and maintained scripts for automated testing and quality assurance.

Technological Unit — **Israel**
Mechanical Designer — *2010–2014*
- Developed mechanical devices, including ideation, 3D CAD design, technical drawings, assembly, testing and documentation.

Leadership and Service

PhD Student Representative, Gender Equality Program, MPI-IS (2022–2024): Collaborated with institute leadership to implement gender equality and diversity initiatives.

Committee Member, Athena Group, MPI-IS (2023–2025): Organizing monthly lunches at the Tübingen site to discuss and promote diversity and inclusion.

Awards and Honors

Top Reviewer: Neurips 2025

Dean's Honor List, The Hebrew University of Jerusalem (MSc): 2021

Weizmann Institute *Amos de-Shalit Summer School* for excellent students in Physics: 2017

Excellent Students in Computer Science Program (BSc): 2016–2018

Technical Skills

Languages: Python, MATLAB, C, C++

Libraries: PyTorch, Scikit-learn

Tools: Git, Jupyter